LINUX ADMINISTRATION LAB SHEET – 3

Lab Setup

```
shivangini@ubuntu:~$ mkdir -p ~/linux_lab3
shivangini@ubuntu:~$ cd ~/linux_lab3
shivangini@ubuntu:~/linux_lab3$ pwd
shivangini@ubuntu:~/linux_lab3$ mkdir -p data backups logs output
shivangini@ubuntu:~/linux_lab3$ touch data/names.txt data/students.txt data/notes.txt
shivangini@ubuntu:~/linux_lab3$ printf 'Zara\nAmit\nNeha\nRavi\nAmit\n' > data/names.txt
shivangini@ubuntu:~/linux_lab3$ printf '101,Amit,85\n102,Neha,92\n103,Ravi,68\n104,Zara,76\n' >
data/students.txt
shivangini@ubuntu:~/linux_lab3$ printf 'Linux shell scripting\nVariables and loops\nPipes and redirection\n' >
data/notes.txt
shivangini@ubuntu:~/linux_lab3$ touch backups/db_01.bak backups/db_02.bak
shivangini@ubuntu:~/linux_lab3$ find . -maxdepth 2 -type f -print
```

Q1. Hello Script

```
#!/bin/bash

name="shivangini"
echo "Hello, $name!"
echo "Welcome to Ubuntu Linux."
shivangini@ubuntu:~/linux_lab3$ nano hello.sh
shivangini@ubuntu:~/linux_lab3$ chmod +x hello.sh
shivangini@ubuntu:~/linux_lab3$ ./hello.sh
shivangini@ubuntu:~/linux_lab3$ bash hello.sh
```

Q2. User Input Script

```
#!/bin/bash

echo "Enter your name:"
read name

echo "Enter your branch:"
read branch

echo "Enter your semester:"
read semester

echo "Welcome, $name!"
echo "Branch: $branch"
echo "Semester: $semester"
shivangini@ubuntu:~/linux_lab3$ nano user_input.sh
shivangini@ubuntu:~/linux_lab3$ chmod +x user_input.sh
shivangini@ubuntu:~/linux_lab3$ ./user_input.sh
```

Q3. Arithmetic Calculator

```
#!/bin/bash

echo "Enter first number:"
read a

echo "Enter second number:"
read b
```

```
echo "Sum: $((a + b))"
echo "Difference: $((a - b))"
echo "Product: $((a * b))"

if [ "$b" -ne 0 ]; then
   echo "Quotient: $((a / b))"
else
   echo "Division by zero is not allowed."
fi
shivangini@ubuntu:~/linux_lab3$ nano calculator.sh
shivangini@ubuntu:~/linux_lab3$ chmod +x calculator.sh
shivangini@ubuntu:~/linux_lab3$ ./calculator.sh
```

Q4. Command Substitution

```
#!/bin/bash

current_date=$(date)
current_directory=$(pwd)
username=$(whoami)

echo "Current Date: $current_date"
echo "Current Directory: $current_directory"
echo "Username: $username"
shivangini@ubuntu:~/linux_lab3$ nano system_info.sh
shivangini@ubuntu:~/linux_lab3$ chmod +x system_info.sh
shivangini@ubuntu:~/linux_lab3$ ./system_info.sh
```

Q5. Number Classification

```
#!/bin/bash

echo "Enter an integer:"
read number

if [ "$number" -gt 0 ]; then
   echo "The number is positive."
elif [ "$number" -lt 0 ]; then
   echo "The number is negative."
else
   echo "The number is zero."
fi
shivangini@ubuntu:~/linux_lab3$ nano number_type.sh
shivangini@ubuntu:~/linux_lab3$ chmod +x number_type.sh
shivangini@ubuntu:~/linux_lab3$ ./number_type.sh
```

Q6. Even or Odd

```
#!/bin/bash

if [ "$#" -ne 1 ]; then
   echo "Usage: ./even_odd.sh number"
   exit 1
fi

number=$1

if [ $((number % 2)) -eq 0 ]; then
```

```
    echo "$number is even."
else
    echo "$number is odd."
fi
shivangini@ubuntu:~/linux_lab3$ nano even_odd.sh
shivangini@ubuntu:~/linux_lab3$ chmod +x even_odd.sh
shivangini@ubuntu:~/linux_lab3$ ./even_odd.sh 8
shivangini@ubuntu:~/linux_lab3$ ./even_odd.sh 7
```

Q7. File and Directory Tests

```
#!/bin/bash

if [ "$#" -ne 1 ]; then
    echo "Usage: ./file_check.sh path"
    exit 1
fi

path=$1

if [ -f "$path" ]; then
    echo "$path is a regular file."

    if [ -x "$path" ]; then
        echo "The file is executable."
    else
        echo "The file is not executable."
    fi

elif [ -d "$path" ]; then
    echo "$path is a directory."
else
    echo "$path does not exist or is neither a regular file nor a directory."
fi
shivangini@ubuntu:~/linux_lab3$ nano file_check.sh
shivangini@ubuntu:~/linux_lab3$ chmod +x file_check.sh
shivangini@ubuntu:~/linux_lab3$ ./file_check.sh data
shivangini@ubuntu:~/linux_lab3$ ./file_check.sh data/names.txt
```

Q8. For Loop Practice

```
#!/bin/bash

if [ "$#" -ne 1 ]; then
    echo "Usage: ./multiplication.sh number"
    exit 1
fi

number=$1

for i in {1..10}
do
    echo "$number x $i = $((number * i))"
done
shivangini@ubuntu:~/linux_lab3$ nano multiplication.sh
shivangini@ubuntu:~/linux_lab3$ chmod +x multiplication.sh
shivangini@ubuntu:~/linux_lab3$ ./multiplication.sh 5
```

Q9. Loop Through Files

```
#!/bin/bash

for file in data/*.txt
do
   if [ -f "$file" ]; then
      echo "Text file found: $file"
   fi
done
shivangini@ubuntu:~/linux_lab3$ nano list_txt.sh
shivangini@ubuntu:~/linux_lab3$ chmod +x list_txt.sh
shivangini@ubuntu:~/linux_lab3$ ./list_txt.sh
```

Q10. While Loop

```
#!/bin/bash

count=10

while [ "$count" -ge 1 ]
do
   echo "$count"
   count=$((count - 1))
done

echo "Done!"
shivangini@ubuntu:~/linux_lab3$ nano countdown.sh
shivangini@ubuntu:~/linux_lab3$ chmod +x countdown.sh
shivangini@ubuntu:~/linux_lab3$ ./countdown.sh
```

Q11. Script Arguments

```
#!/bin/bash

echo "Script Name: $0"
echo "First Argument: $1"
echo "Second Argument: $2"
echo "Number of Arguments: $#"
echo "All Arguments: $@"
shivangini@ubuntu:~/linux_lab3$ nano arguments.sh
shivangini@ubuntu:~/linux_lab3$ chmod +x arguments.sh
shivangini@ubuntu:~/linux_lab3$ ./arguments.sh shivangini MCA
shivangini@ubuntu:~/linux_lab3$ ./arguments.sh Linux Ubuntu
```

Q12. Exit Codes

```
#!/bin/bash

if [ "$#" -ne 1 ]; then
   echo "Usage: ./status.sh directory"
   exit 1
fi

if [ -d "$1" ]; then
   echo "Directory exists."
   exit 0
else
   echo "Directory does not exist."
   exit 1
fi
```

```
shivangini@ubuntu:~/linux_lab3$ nano status.sh
shivangini@ubuntu:~/linux_lab3$ chmod +x status.sh
shivangini@ubuntu:~/linux_lab3$ ./status.sh data
shivangini@ubuntu:~/linux_lab3$ echo $?
shivangini@ubuntu:~/linux_lab3$ ./status.sh unknown
shivangini@ubuntu:~/linux_lab3$ echo $?
```

Q13. Output Redirection

```
shivangini@ubuntu:~/linux_lab3$ ls data > output/files.txt
shivangini@ubuntu:~/linux_lab3$ cat output/files.txt

shivangini@ubuntu:~/linux_lab3$ echo "First entry" > output/files.txt
shivangini@ubuntu:~/linux_lab3$ echo "Second entry" >> output/files.txt
shivangini@ubuntu:~/linux_lab3$ cat output/files.txt
```

Q14. Append a Log

```
#!/bin/bash

echo "Enter your message:"
read message

echo "$(date '+%Y-%m-%d %H:%M:%S') - $message" >> logs/activity.log

echo "Log entry saved successfully."
shivangini@ubuntu:~/linux_lab3$ nano log_entry.sh
shivangini@ubuntu:~/linux_lab3$ chmod +x log_entry.sh
shivangini@ubuntu:~/linux_lab3$ ./log_entry.sh
shivangini@ubuntu:~/linux_lab3$ cat logs/activity.log
```

Q15. Input Redirection and Sorting

```
shivangini@ubuntu:~/linux_lab3$ sort < data/names.txt
shivangini@ubuntu:~/linux_lab3$ sort -u < data/names.txt
```

Q16. Error Logging

```
shivangini@ubuntu:~/linux_lab3$ ls missingfile.txt 2> logs/errors.log
shivangini@ubuntu:~/linux_lab3$ cat logs/errors.log
```

Q17. Combined Output and Error Logging

```
shivangini@ubuntu:~/linux_lab3$ ls data missingfile.txt > logs/combined.log 2>&1
shivangini@ubuntu:~/linux_lab3$ cat logs/combined.log
```

Q18. Pipeline Practice

```
shivangini@ubuntu:~/linux_lab3$ ps aux | grep bash
shivangini@ubuntu:~/linux_lab3$ ps aux | grep bash | wc -l
```

Q19. Backup Checker Script

```
#!/bin/bash

if [ "$#" -ne 1 ]; then
    echo "Usage: ./backup_check.sh directory"
```

```
    exit 1
fi

directory=$1

if [ ! -d "$directory" ]; then
    echo "Directory does not exist."
    exit 1
fi

count=0

for file in "$directory"/*.bak
do
    if [ -f "$file" ]; then
        echo "Backup found: $file"
        count=$((count + 1))
    fi
done

echo "Total backup files: $count"

if [ "$count" -eq 0 ]; then
    echo "No backup files found" >> logs/backup_log.txt
fi
shivangini@ubuntu:~/linux_lab3$ nano backup_check.sh
shivangini@ubuntu:~/linux_lab3$ chmod +x backup_check.sh
shivangini@ubuntu:~/linux_lab3$ ./backup_check.sh backups
shivangini@ubuntu:~/linux_lab3$ ./backup_check.sh data
shivangini@ubuntu:~/linux_lab3$ cat logs/backup_log.txt
```

Q20. Integrated File Report

```
#!/bin/bash

if [ "$#" -ne 1 ]; then
    echo "Usage: ./file_report.sh filename"
    exit 1
fi

file=$1

if [ ! -f "$file" ]; then
    echo "Error: File does not exist."
    echo "$(date '+%Y-%m-%d %H:%M:%S') - File not found: $file" >> logs/execution_history.log
    exit 1
fi

if [ ! -r "$file" ]; then
    echo "Error: File is not readable."
    echo "$(date '+%Y-%m-%d %H:%M:%S') - File not readable: $file" >> logs/execution_history.log
    exit 1
fi

echo "File: $file"
echo "Line, word and character counts:"
wc "$file"

echo "$(date '+%Y-%m-%d %H:%M:%S') - Successfully processed: $file" >> logs/execution_history.log
```

```
echo "Processing record saved."
shivangini@ubuntu:~/linux_lab3$ nano file_report.sh
shivangini@ubuntu:~/linux_lab3$ chmod +x file_report.sh
shivangini@ubuntu:~/linux_lab3$ ./file_report.sh data/names.txt
shivangini@ubuntu:~/linux_lab3$ ./file_report.sh unknown.txt
shivangini@ubuntu:~/linux_lab3$ cat logs/execution_history.log
```